Machine Learning

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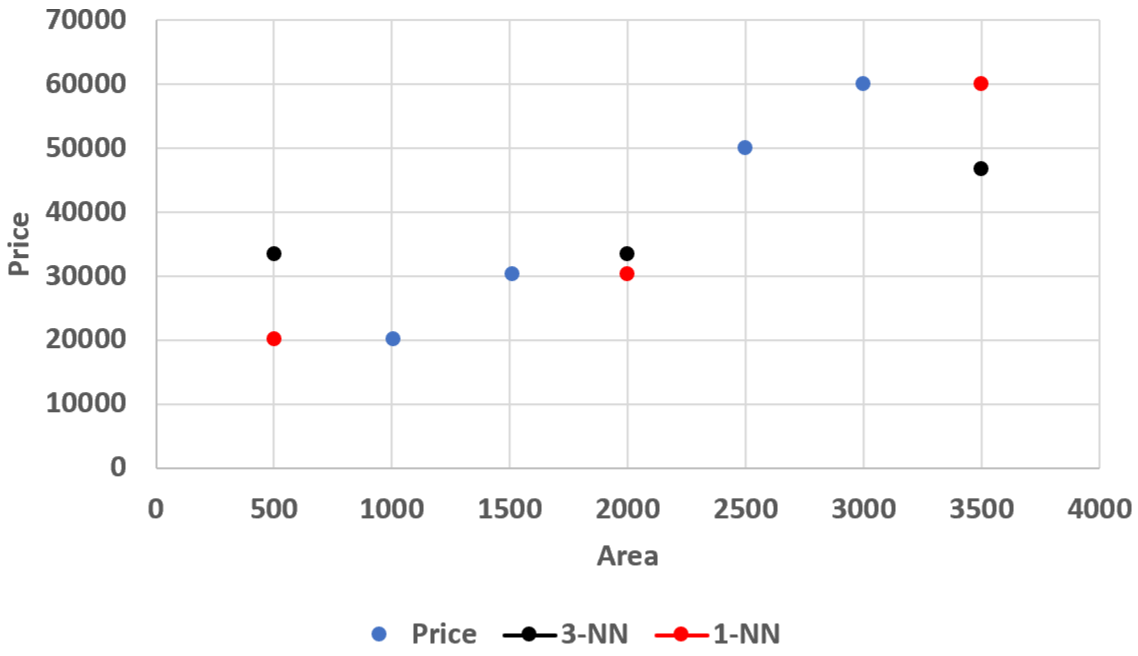
Lecture 09: Linear Regression

Find the price!

Area	Price
1,510	30,250
1,005	20,150
2,500	50,050
3,000	60,050

Price for Area: 500? 2000?, 3500?

Area	1-NN	3-NN
500	20,150	30,150
2,000	30,250	33,483.33
3,500	60,050	50,050



Linear Regression

https://www.desmos.com/calculator/4bshz7dwkw

Variance and Standard Deviation

- Variance or standard deviation tells us how much a (single) variable/quantity varies w.r.t. its mean 'OR' how spread out the data is around the center of the distribution (the mean).
- The value of variance doesn't give us any information, so we take its square root to calculate the Standard Deviation .
- **Low standard deviation** implies that most values in the dataset are **close to the mean.**
- **High standard** deviation implies that the values in the dataset are more **broadly spread out.**

$$\boldsymbol{\sigma_x} = \sqrt{\frac{\sum(\boldsymbol{x_i} - \boldsymbol{\bar{x}})^2}{\boldsymbol{n} - \mathbf{1}}}$$

Covariance

- Covariance measures the direction of the relationship between two random variables, and its value range from negative infinity to positive infinity.
- It tells us how two variables/quantities, and are related to each other or how the mean values of two random variables move together.
- It can be calculated using following formula:

$$\operatorname{cov}(x, y) = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

- **Positive covariance** indicates that two variables tend to move in the same direction.
- **Negative covarian**ce indicates that two variables tend to move in the inverse direction.
- **Zero covariance** indicates that two variables have no relationship between each other

Correlation

- Correlation tells us the direction as well as the magnitude of the relationship between two random variables, and the value of correlation coefficient ranges from -1 to +1.
- There are several types of correlation coefficients. The Pearson Correlation Coefficient (developed by Karl Pearson) denoted by R or ρ is the most common and is defined by:

$$R_{x,y} = \frac{cov(x, y)}{\sigma_x \sigma_y}$$

$$R_{x,y} = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2}\sqrt{\sum(y_i - \bar{y})^2}}$$

Correlation

The value of Correlation Coefficient always lie between -1 and +1. We can interpret as follows:

- If corr =1, then Perfect Positive linear relationship exist between two variables.
- If 0.7 <= corr <1, then Strong Positive linear relationship exist between two variables.
- If 0.3 <= corr <0.7, then Moderate Positive linear relationship exist between two variables.
- If -0.3 <= corr <0.3, then Very Week linear relationship exist between two variables.
- If -0.7 <= corr <-0.3, then Moderate Negative linear relationship exist between two variables.
- If -1 < corr <-0.7, then Strong Negative linear relationship exist between two variables.
- If corr = -1 then Perfect Negative linear relationship exist between two variables.

Regression

- Regression is a statistical technique that relates a dependent variable to one or more independent variables.
- It is used to predict/forecast the value of a dependent variable based on the known values of the independent variable(s)
  - If output variable is a real value like salary, cost, gpa etc, then you are predicting present but unknown values.
  - If output variable is a real value like time, then you are forecasting .

Correlation vs Regression

- Regression can differentiate between dependent and independent variables, while correlati cannot.
- The reg(x,y) and reg(y,x) are completely different, while the corr(x,y) and corr(y,x) are identical.
- Regression can measure causality as well, while correlation just tell the direction of movement between two variables.
- Regression can measure the strength and direction of both linear as well as non-linear relationship, while correlation can do this for linear relationship only.
- The graphical representation of a correlation is a single point, while a line visualizes a linear regression.

Types of Regression

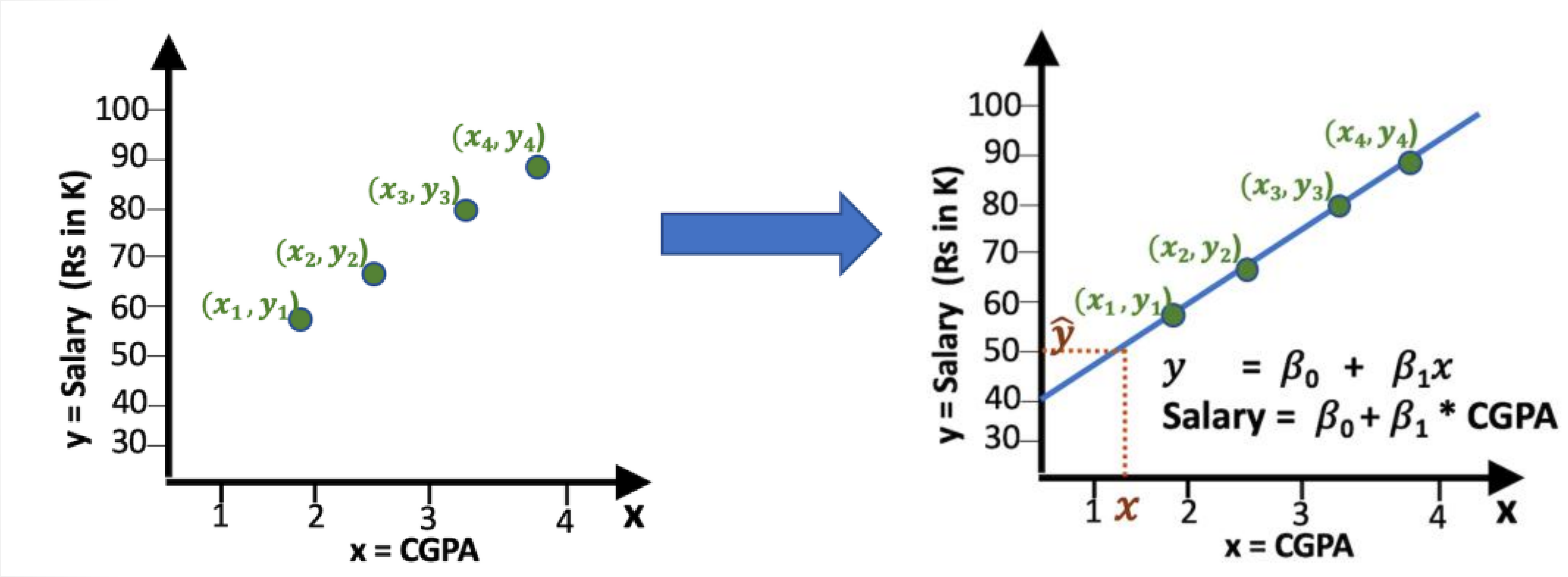
Some common types are:

- Simple Linear Regression
- Multiple Linear Regression
- Polynomial Regression
- Logistic Regression
- Ridge Regression
- Lasso Regression
- Bayesian Linear Regression
- Support Vector Regression (SVR)
- Decision Tree Regression

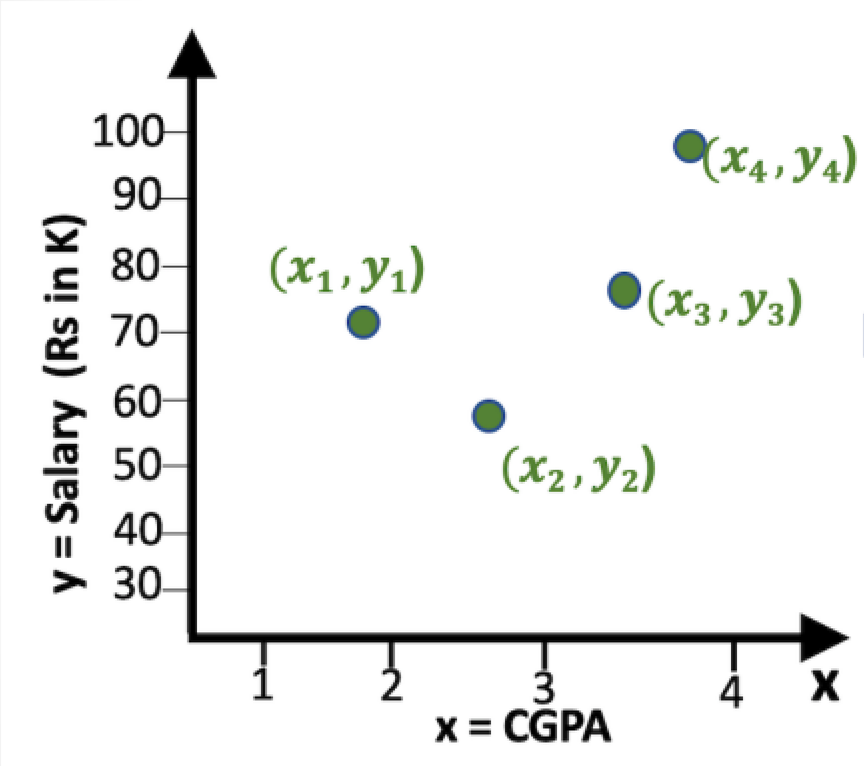
Simple Linear Regression

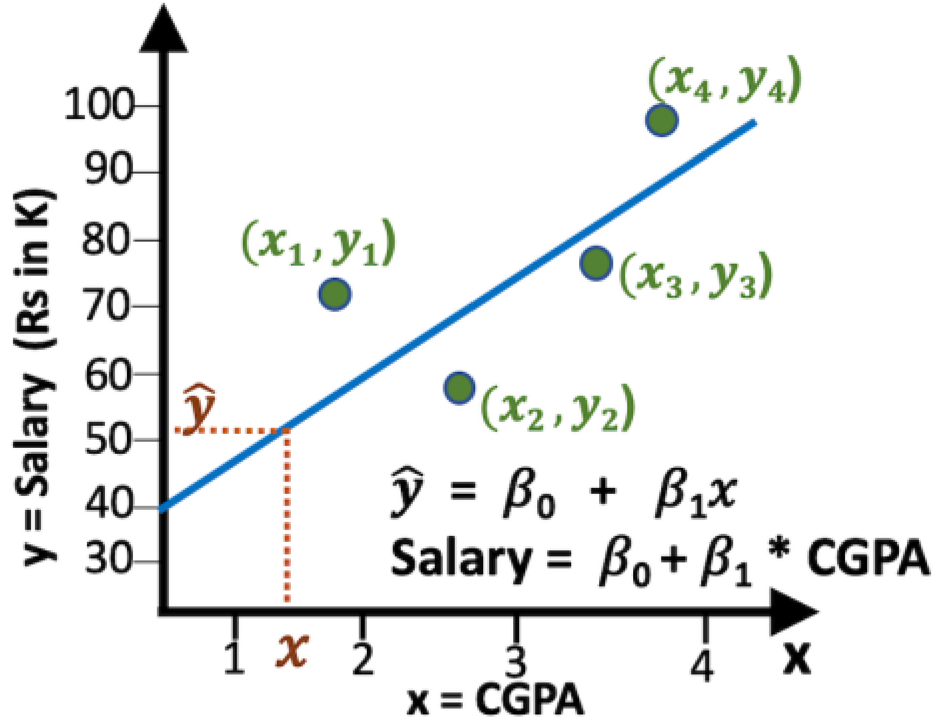
- Linear regression is a type of model where the relationship between a dependent variable and one or more independent variables is assumed to be linear.
- There are two kinds of Linear Regression Model:
  - Simple Linear Regression: A linear regression model with one independent and one dependent
  - variable.
  - Multiple Linear Regression: A linear regression model with more than one independent variable and one dependent variable.

An Intuitive Understanding of Simple Linear Regression



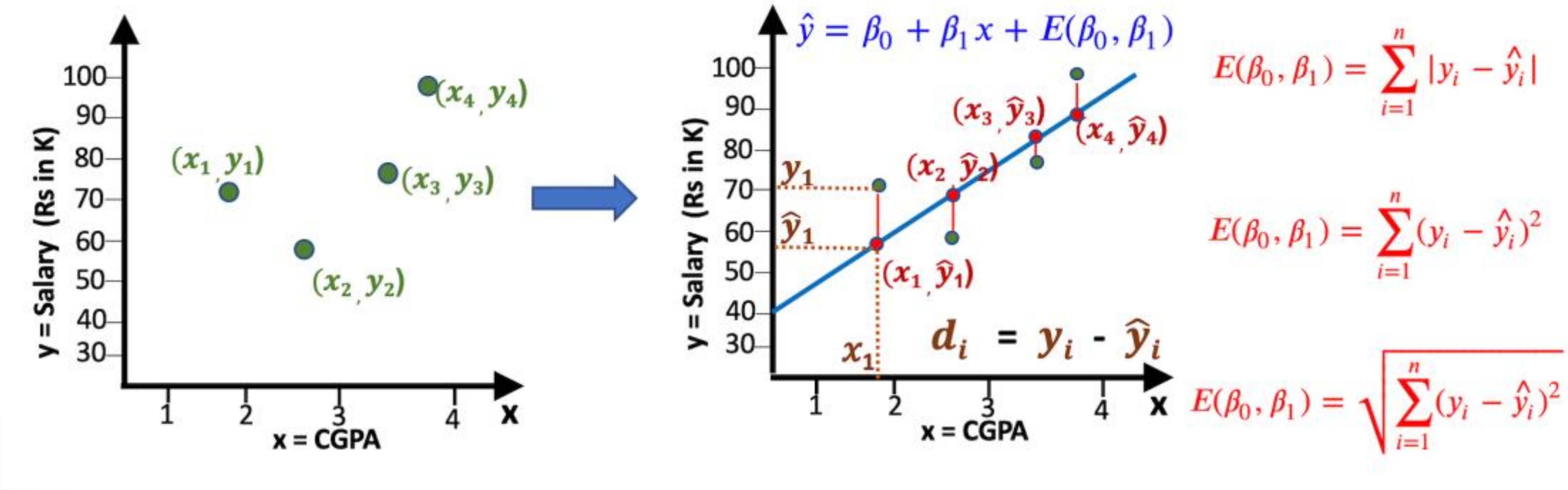
An Intuitive Understanding of Simple Linear Regression



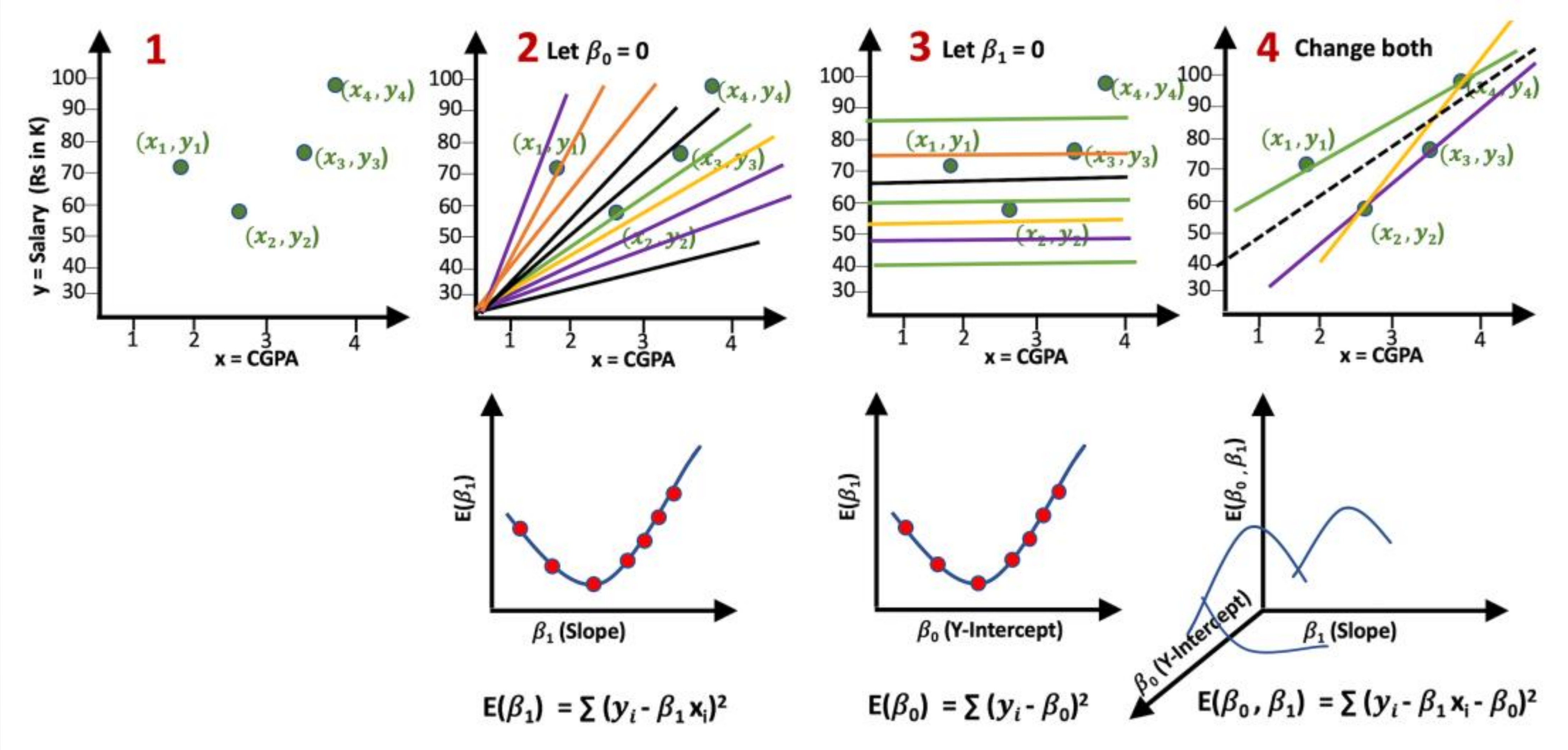


An Intuition of Ordinary Least Squares Method (OLS)

- The process of fitting the best-fit line is called linear regression.
- Ordinary Least Squares is a technique that works by minimizing the sum of squares of the differences between the observed dependent variable (cgpa) and the independent variable (salary).



Minimize the Cost Function to Derive $\beta 0$ and $\beta 1$



Example (Simple Linear Regression using OLS)

- Consider that a teacher has collected a sample data of seven students, their GPA and the number of hours they have studied on daily basis in the entire semester.
- Suppose, a teacher want to determine the relationship between the two variables GPA and study hours is positive or negative.
- The teacher also wants to determine how much impact the dependent variable study hours has on the independent variable GPA
- For this she has collected a sample data of seven students as tupple containing (daily study hours and acquired GPA):

	study_hours	gpa
0	1.0	1.4
1	2.0	1.6
2	3.0	2.5
3	4.0	2.6
4	5.0	3.5
5	6.0	3.7
6	7.0	4.0

Example (Simple Linear Regression using OLS)

- In the case of a model with a single predictor, there is a fairly straightforward linear least squares formula
- we can use to estimate :

$$\hat{\beta}_1 = \frac{(\overline{x_i y_i}) - \bar{x}_i \bar{y}_i}{\overline{x_i^2} - \bar{x}_i^{\,2}}$$

$$\hat{\beta}_1 = \frac{\mathrm{cov}(x,y)}{\sigma_x^2}$$

$$\hat{\beta}_1 = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2}$$

	study_hours	gpa
0	1.0	1.4
1	2.0	1.6
2	3.0	2.5
3	4.0	2.6
4	5.0	3.5
5	6.0	3.7
6	7.0	4.0

Thank You